

SAGNAW COUNTY AP, MI, USA

WMO: 722125

Lat: 43.433N

Lon: 83.867W

Elev: 600

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04829

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	5.3	-7.6	3.7	1.7	-1.8	5.1	7.2	27.0	26.0	24.5	27.0	6.6	270	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	89.7	72.0	86.4	70.7	82.3	68.7	75.2	84.7	73.3	82.0	71.8	79.7	10.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	122.5	79.9	70.5	114.7	77.9	69.7	111.8	77.1	39.0	83.5	37.3	81.9	36.0	79.7	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.2	20.6	18.5	DB	-4.6	93.1	5.2	2.8	-8.3	95.1	-11.4	96.7	-14.3	98.2	-18.1	100.2	(4)
(5)			WB	-5.2	78.5	5.1	1.9	-8.9	79.9	-11.9	81.0	-14.7	82.1	-18.4	83.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	24.6	24.5	34.2	46.2	58.3	67.3	71.0	68.7	62.4	50.9	39.3	29.6		
	DBStd	18.75	10.50	10.29	10.84	8.74	8.88	6.66	5.90	5.44	7.38	8.44	8.62	8.63		
	HDD50	3267	789	714	505	173	20	0	0	0	2	93	338	633		
	HDD65	6698	1253	1134	955	564	252	49	13	25	137	446	772	1098		
	CDD50	2615	1	0	16	61	276	519	652	581	373	120	16	0		
	CDD65	568	0	0	1	1	43	117	200	141	57	8	0	0		
	CDH74	5155	0	0	13	38	471	1089	1890	1092	489	73	0	0		
	CDH80	1456	0	0	3	2	124	322	615	257	119	14	0	0		
(14) Wind	WSAvg	8.2	9.5	9.5	9.2	9.9	8.4	7.3	6.3	5.9	6.4	8.2	9.0	9.2		
(15) Precipitation	PrecAvg	31.5	1.9	1.7	1.9	3.0	3.5	3.4	3.2	3.3	2.8	2.9	2.6	2.0		
	PrecMax	40.0	3.6	4.6	4.8	6.7	7.8	7.3	4.9	5.1	5.7	5.6	5.5	4.4		
	PrecMin	25.6	0.2	0.3	0.3	1.2	1.4	1.2	0.7	1.3	0.4	0.9	0.7	0.6		
	PrecStd	4.0	0.9	1.1	1.0	1.5	1.6	1.4	1.1	1.1	1.5	1.3	1.5	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	55.3	73.2	79.2	88.2	91.1	93.1	90.3	88.4	81.5	68.1	56.8		
		MCWB	52.5	49.3	61.2	61.1	69.4	72.0	74.7	72.0	70.3	68.7	57.3	53.1		
	2%	DB	47.8	47.8	63.2	72.9	82.4	88.0	90.1	86.2	82.4	74.8	61.2	49.5		
		MCWB	45.5	43.9	54.2	57.0	66.1	71.0	73.2	71.9	69.1	62.5	53.1	46.8		
	5%	DB	41.4	41.2	55.3	68.1	79.5	83.6	86.5	82.0	79.4	70.1	56.6	44.8		
		MCWB	38.1	37.2	48.8	55.1	63.8	70.1	71.5	69.4	66.6	61.9	50.2	42.3		
	10%	DB	36.8	36.8	50.4	62.8	74.7	80.7	82.2	80.6	75.5	64.4	53.5	40.6		
		MCWB	35.3	34.5	44.4	51.6	61.4	66.4	69.1	68.2	64.9	57.9	48.8	37.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	51.7	62.8	64.3	72.4	75.8	78.1	77.0	74.7	70.9	59.4	54.6		
		MCDB	53.7	55.7	75.0	73.4	82.4	87.4	88.7	84.4	83.0	79.8	63.9	56.5		
	2%	WB	45.9	44.2	55.9	59.8	69.5	73.1	75.9	73.9	72.0	66.7	54.5	46.6		
		MCDB	47.1	48.8	61.2	69.1	77.0	83.3	85.8	82.1	79.4	72.0	59.5	48.0		
	5%	WB	39.1	38.0	50.7	56.4	67.0	71.3	73.9	72.1	69.6	62.9	51.4	42.3		
		MCDB	41.8	40.8	56.1	66.0	76.2	79.5	82.4	79.7	75.4	67.1	56.4	44.6		
	10%	WB	35.5	34.7	44.2	52.8	64.2	69.3	72.1	70.5	67.4	59.0	47.7	37.5		
		MCDB	37.1	37.2	49.4	61.9	72.7	77.2	79.9	76.8	74.3	64.9	51.9	39.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	13.8	15.9	19.9	21.0	20.2	20.5	19.6	20.8	17.4	13.4	10.0		
		MCDBR	15.4	15.9	24.6	28.5	25.0	22.9	23.4	21.5	23.8	24.7	20.6	14.9		
	5% WB	MCWBR	13.8	13.0	17.0	16.4	12.6	11.3	10.7	10.5	12.1	14.5	14.6	13.1		
		MCWBR	15.1	15.6	23.2	25.3	22.2	19.5	19.7	17.9	20.1	20.8	18.7	14.9		
(40) Clear-Sky Solar Irradiance	taub		0.292	0.314	0.335	0.369	0.394	0.422	0.411	0.403	0.376	0.346	0.312	0.288		
	taud		2.380	2.321	2.375	2.306	2.293	2.245	2.289	2.302	2.374	2.449	2.490	2.442		
	Ebn at Noon		266	279	287	284	279	270	272	270	270	265	258	258		
	Edh at Noon		25	31	34	39	41	43	41	39	34	27	22	21		
(44) All-Sky Solar Radiation	RadAvg		447	728	1131	1452	1706	1919	1930	1647	1313	825	502	355		
	RadStd		45	93	119	124	150	107	93	70	125	81	64	31		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page